How to select the model of Thermotechnical Treasure (Thermotechnical Multi-function Calibrator) and HART Handheld Communicator for on-site use

In the field of industrial thermal automation, during on-site debugging, maintenance, and calibration, the AILEIKE Thermal Engineering Treasure (thermal engineering multi-function calibrator) and the HART handheld operator are typical representatives of instruments used in industrial sites. Both have a handheld and portable design, but differ in functional principles and application scenarios.

The difference between the Thermotechnical Treasure and the HART handheld communicator lies in that the Thermotechnical Treasure is a universal "signal generator and calibrator", while the HART handheld communicator is an "instrument debugger". Simply put, one focuses on the measurement and output of "physical signals", while the other focuses on the configuration and diagnosis of "digital communication".

1. Basic essential differences

1. AILEIKE Thermal Engineering Handbook (Representative Model: ALKH512)

Thermal Engineering Treasure is a thermal signal simulation + measurement calibrator, which belongs to the category of "metrological calibration tools". Its function is to actively output standard signals and collect on-site measurement signals, used to verify the accuracy of various thermal instruments and control systems. It serves as a "standard signal source + calibrator" on the thermal engineering site.

2. AILEIKE HART handheld device (representative model: ALK475TREX series)

The HART handheld communicator is a HART protocol instrument communication debugger, categorized as a "digital debugging tool". Its primary function is to communicate with HART devices such as transmitters and flowmeters, allowing for the reading of device parameters, modification of measurement ranges, zero resetting, configuration, and fault diagnosis. It does not output standard calibration signals, nor is it responsible for instrument accuracy calibration.

II. Functional differences

1. Thermal Engineering Handbook: Calibration, Signal Measurement, and Output

The Thermotechnical Treasure Book is used for signal measurement and output (such as thermocouple/resistance signals, outputting 0~24mA current) and serves as a field calibration tool.

Signal output function: It can output various thermal standard signals such as voltage, current, resistance, thermocouple, frequency, and loop power supply.

Signal measurement function: It can measure voltage, current, thermal resistance, thermocouple, and frequency, collect output signals from on-site instruments and sensors, and compare them with standard values to determine instrument errors.

Thermal Engineering Handbook Inquiry: Equipped with built-in temperature scale standard data, it allows for quick inquiry of temperature parameters corresponding to thermocouples and thermal resistors, enabling on-site parameter verification and data conversion.

Adaptation and verification objects: thermal calibration equipment such as temperature transmitters, thermocouples, thermal resistors, digital display instruments, and flow meters.

2. HART handheld communicator: communication, configuration and debugging

The HART handheld communicator is used for instrument communication, modifying ranges, reading diagnostic information, etc. It is a "digital debugging tool".

Device communication identification: It can communicate with domestic and foreign HART instruments such as 1151, 3051, and EJA, and is compatible with HART275, HART375, and HART475 series protocols.

Adjustable with transmitter: zero point, unit, user range, pressure value, current, damping, output mode, loop test, workstation number, date, descriptor, information, final assembly number, equipment test.

Parameter reading, writing, and modification: Real-time reading of device process variables, current values, percentage range, device model, serial number, and fault codes; ability to modify parameters such as instrument range, zero point, damping, unit, output mode, etc.

Fault diagnosis and maintenance: Troubleshoot communication faults and parameter anomalies of HART devices such as transmitters and electromagnetic flowmeters, enabling on-site debugging without disassembly.

Switching between Chinese and English: compatible with various imported and domestic HART instruments, with strong operational adaptability.

Compatible with: Automation instruments with HART protocol.

3. Both the Thermal Engineering Handbook and the HART handheld communicator are equipped with 24VDC power supply

It can directly supply power to on-site transmitters, sensors, etc., without requiring an external power source, meeting the needs of instrument debugging in power-free scenarios.

III. Working Principle and Usage Scenario

1. Different working principles

Thermal Engineering Treasure Book: Sending/receiving physical signals (voltage, current, resistance) through probe wires.

HART handheld communicator: By connecting in parallel to the instrument loop, it superimposes digital signal communication on the 4-20mA signal without interrupting the existing control.

2. Different on-site usage scenarios

Thermal Engineering Treasure Book: Annual instrument inspection and calibration, new equipment commissioning, troubleshooting of large instrument errors, maintenance of ordinary instruments without protocols, and on-site temporary power supply commissioning are essential tools for thermal engineering measurement, calibration, and inspection.

HART handheld communicator: It allows for range modification, zero point fine adjustment, device parameter reset, and unified debugging of batch instrument parameters for instruments with HART protocol. It is used for parameter configuration, such as adjusting transmitter ranges and viewing fault information.

IV. How to select and use

1. If the work requires adjusting the transmitter range, clearing the zero point, troubleshooting HART protocol instrument faults, or modifying device parameters, a HART handheld device can be used with a 24V power

supply function. However, the Thermal Engineering Handbook does not have a communication configuration function;

2. Standard combination for industrial thermal engineering sites: Thermal Engineering Treasure + HART handheld device, covering two major functional areas of "signal measurement and output" and "configuration", suitable for instrument debugging and maintenance in various scenarios.